

Post-detection and SETI in Society: developments in 2024



(IAA SETI Committee Members: please see full written report for details, links, etc.)

Update for IAA SETI Committee, 16 Oct 2024, at IAC Milan

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General Status

- Many committee members are interested in post-detection / discovery & society, although at present we aren't formally organized into a PD Sub-Committee within the Committee. Perhaps this will change by the next update.
- Meanwhile, considerable work and discussion relevant to PD continues.
- In late 2022 a small task group started work specifically on Protocols revisions. Today we are presenting a Draft Revised *"Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)"* to the Committee for discussion.
- Interest is intensifying in "discovery and society" issues for ETL and ETI for good reasons, including the proliferation of potential discovery types and an increasingly complicated and risky communication landscape. However, funding structures have not kept pace.

Overview

- Abstracts related to Post-detection in 2024 IAC SETI sessions
- Some relevant recent and upcoming activities of SETI Committee members, including several I'll review in a bit more detail:
 - IAU/Kavli symposium (IAUS 387): *(Toward) Discovery of Life Beyond Earth and its Impact*
 - International Institute of Space Law (IISL) & Law Working Group
 - SETI Post-Detection Hub
 - NASA *Communicating Discoveries in the Search for Life in the Universe* workshop
- The UN, Foresight/Futures, and Space
- The Protocols: a Draft Revised "Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)" - L. Tennen will present details, just two comments from me:
 - Potential future Best Practices Guidelines, and Post-Detection subcommittee resurrection
 - A note on history and the need for preservation/access

Abstracts related to Post-Detection at IAC 2024

<https://iafastro.directory/iac/browse/IAC-24/A4/>

- **Daniela De Paulis *et al***, KEYNOTE: "Billingham Cutting-Edge Lecture" - Global outreach and cultural impact of A Sign in Space, an interdisciplinary simulation of a First Contact scenario. [Abstract](#)
- **Chelsea Haramia**, Causal Impotence and Cosmic Messaging: A Logical Response to the Barn Door Argument. [Abstract](#)
- **Paolo Musso *et al***, The "Toulouse Reboot": Opening a New Era in the Studies on Interstellar Messages. [Abstract](#)
- **Andrea Harrington & Charles Stotler**, State Responsibility for First Contact Under International Law. [Abstract](#)
- **Les Tennen *et al***, The Future of the SETI Post-Detection Protocols: Progress Towards Revisions. [Abstract](#)
- **Raoul Cardellini**, Assessing the consistency between SETI and metalaw and the collective interest of humankind in the exploration and use of outer space. [Abstract](#)
- **Julia DeMarines and Chelsea Haramia**, An Active and Community-Based Approach to Inclusive Debates in SETI Ethics. [Abstract](#)

Abstracts related to Post-Detection at IAC 2024, cont'd

- **Lorenzo De Piccoli**, Technosignatures. The 400-Year Search for Extraterrestrial Artefacts. [Abstract](#)
- **Giorgio Gaviraghi**, Artificial intelligences (AIs) could potentially serve as the catalyst for the first contact between our society and an alien one. [Abstract](#)
- **Kate Genevieve *et al***, Plurality in Post Detection Scenarios. [Abstract](#)
- **Nigar Ismayilzada *et al***, Ethical frontiers and social echoes: navigating the implications of SETI research. [Abstract](#)
- **Ali Mammadov**, Discovering Extraterrestrial Life in the Universe: Scientific, Technical, and Interdisciplinary Dimensions of Extraterrestrial Life and SETI's contributions to finding Extraterrestrial Life. [Abstract](#)
- **Samiksha Raviraja *et al***, The Discovery of Alien Life and How That could Impact Fundamental Human Beliefs, Societal Views and Philosophies. [Abstract](#)
- **Gabriela Radulescu**, The Relational Legacy of Extraterrestrial Intelligence. [Abstract](#)

Some PD-relevant activities of SETI Committee members, recent and upcoming

N.b. This compilation is not comprehensive. Please do share your updates/information with me.

- Continued public outreach by SETI Institute and Breakthrough Listen, continued public engagement by other SETI projects
- Daniela de Paulis and team, [A Sign in Space](#)
- PSETI: [seminar series](#), [OotO](#), and [PSETI Symposium 2024](#)  Hopefully a PD breakout and talks again
- [AbSciCon 2024](#), May, including large “Social Sciences, Humanities, and Astrobiology” session, and much discussion about communicating astrobio science / discoveries to public
- [Technosignatures Seminar Series](#) at NASA Goddard
- [“Breaking News: We Found Extraterrestrial Life!”](#) workshop at the Lorentz Centre, Leiden U., Sept 2024 (linked to NASA *Communicating Discoveries* workshop). White paper to come.
- [Astrobiology and the Future of Life](#), NASA Astrobiology symposium, Oct 2024
- Also substantial involvement from Committee members in (next pages): SETI Post-Detection Hub, IISL SETI & Law Group, IAU/Kavli Symposium, NASA *Communicating Discoveries* workshop

SETI Post-Detection Hub

- Coordinator: John Elliott john.elliott@st-andrews.ac.uk , longtime SETI Committee member
- a [UKSRN](#) initiative, is jointly hosted by the [St Andrews Centre for Exoplanet Science](#) (led by Committee member Martin Dominik and Emily Finer), the Centre for Global Law and Governance, the Durham Law School, and the Jodrell Bank Centre for Astrophysics of U. Manchester
- Membership has grown since Nov 2022 launch: now  researchers (fairly even split UK/international), including 12 members of the IAA SETI Committee.  have worked on SETI topics for many years, others are newer to the subject but bring significant interdisciplinary expertise e.g. sci comm research, public policy, risk, etc. Collaborations are coalescing: reading groups and regular meetings help.
- Current endeavours include: funding applications (for projects and for the Hub as a centre); student networks recently established; participation in British and Irish Science Festivals; talks (e.g. many at IAU/Kavli); discussions with UK government; media (filming, interviews, articles)
- Scenarios work over the past year (G. Profitiliotis leading): a tool from Futures Studies/strategic foresight for thinking about possible futures 5-25 years from now, and building *futures literacy*. A future SETI detection would take place in a different social/technological world, effectively.

International Institute of Space Law (IISL) Working Group on SETI and Law

- Call for members went out via IISL in Dec 2023, initiated by [Michael Bohlander](#) (Durham).
- Inaugural meeting Feb 2024, including some IAA SETI Committee members (M. Garrett, L.I. Tennen, J. Elliott, J. Wright, and K. Denning), SETI community members G Profitiliotis, H Neilson, Gabriela Radulescu, and legal scholars (some new to SETI, others with some background), ~20 members
- Many conversations among members Feb – Sept 2024 about foundations, directions etc. General remit: legal issues that would follow a confirmed detection of ETI, with actual or potential communication. Challenges include interdisciplinary conversations, background resources, and bridging different preconceptions about: ETI; realistic discovery situations; and likelihood of mutual intelligibility. Details of the WG's focus and intended outputs are evolving.
- As of Sept 2024, the Working Group will go forward under the leadership of [Andrea Harrington](#). It is scheduled to run until IAC 2026. More information next year. Contact Harrington for more info.

IAU/Kavli Symposium, “(Toward) Discovery of Life Beyond Earth and its Impact”

- 15-19 April 2024, Durham UK, in-person and virtual. SOC Chair: Hermine Landt.
- in-depth five-day [program](#) with speakers from many fields, including astrobiology, astronomy, law and policy, media, social sciences, philosophy, theology, and futures studies. [Abstracts](#).
- Multiple members of the IAA SETI Committee were co-organisers of the Symposium (including C. Oliver, J. Haqq-Misra, M. Dominik), or presenters (L. Tennen, J. Elliott, J. Rummel, D.  is, K. Denning, S. Croft, S. Webb). Multiple PD Hub members too. 120 attendees, from all five continents.
- Would have been ideal to have more SETI astronomers present; sched conflict with SETI Inst symposium
- Interdisciplinary discussions about discovery processes, evidence, uncertainty, consensus, Protocols, policy, science communication, science fiction; multiple case studies from key players e.g. Venus phosphine; short futures-studies-driven exercises playing through a discovery situation; strong contributions from media/communication experts
- Proceedings forthcoming: Cambridge University Press in 2025 (eds H. Landt, C. Oliver and M. Dominik)
- Meeting report: Nováková, J.N. Toward discovery of life beyond Earth and its impact. *Nat Astron* **8**, 799–800 (2024). <https://doi.org/10.1038/s41550-024-02320-w>
- Questions? Committee members C. Oliver, M. Dominik, J. Haqq-Misra can tell you more

NASA Astrobio Program workshop “Communicating Discoveries in the Search for Life in the Universe”

- All virtual, Feb-Mar 2024 (3 sessions), hundreds of applicants, two tiers (Tier 2: just materials & Slack)
 - Tier 1 active participants in workshop sessions + Slack: 82 people = 31 astrobios, 14 sci journalists, 14 academics; 9 sci comm researchers; 5 sci content creators; 9 other. Many do cross-cutting work. Balance of expertise, career stage, backgrounds. Many with direct experience. Several IAA SC members & technosig researchers.
- Content: background summaries on history of astrobiology, science communication, and social science and astrobiology; case studies (ALH 84001; GFAJ-1; LIGO gravity waves; climate change; COVID 19); possible future discovery situations (exoplanet biosig remote detection; Enceladus in situ detection; MSR)
- Format: small group breakout sessions rather than formal talks, i.e. emphasis on connecting different perspectives; additional meetings after the workshop
- Concrete recommendations: careful communication plans; more research on optimal communication strategies especially re: scientific uncertainty and prolonged verification; scientists want more media training and resources; handling mis/disinformation is an expert skill; need partnerships between scientists, social scientists, humanities scholars, communication experts; fund more research on societal reactions; ensure researcher safety; institutions need to step up their support, beyond usual press office practices; integrate interdisciplinary research into science programs on an ongoing basis. **Of clear relevance to the technosignatures / post-detection community.**
- Co-written workshop report (now in feedback/editing phase) to be published in *Astrobiology* in 2025
- Article, Sarah Scoles, “[If Alien Life Is Found, How Should Scientists Break the News?](#)”, Sci Am.

NASA and “Communicating Discoveries”

A connected, very recent development is the NASA Science Mission Directorate [Request for Information of 8 Oct 2024](#), as an initial stage in developing the 2025 NASA Decadal Astrobiology Research and Exploration Strategy. In addition to inviting research community input on a suite of science topics, the SMD is asking for feedback on:

- *“Topic 8: Prepare for the Discovery of Life Beyond Earth and Subsequent Post-Discovery Activities. Recognize that finding evidence of life beyond Earth, whether present or extinct, represents a pivotal evolution in the field of astrobiology. In tandem with community efforts, focused on refining the science and mission requirements for life detection, e.g., the Search for Life Mission Analysis Group, SFL-SAG, should be the development of frameworks for international cooperation and public engagement to responsibly handle such discoveries. Build on previous efforts from recent workshops e.g., “Communicating Discoveries in the Search for Life in the Universe” and prepare for activities – including post-discovery science – that consider and respond to the profound implications of life’s existence elsewhere, ensuring the field evolves to address new scientific, ethical, and societal changes.”*
- Submit your white papers! (deadline Feb 2025) and stay tuned...

United Nations, Foresight/Futures, Space

Thanks to Dr George Profitiliotis, UN
Development Programme Futures
Fellow in 2024, for contributions & info.

- The SETI Committee has aspired to interest the UN in SETI since the 1970s (Rudolf Pešek, first Chair)
- This year, from UNDP Strategy and Futures Team of the Executive Office of the Administrator:
 - [Signals Spotlight 2024](#) annual report: *“if and when extraterrestrial life were discovered, it would require new forms of inter-species interaction and communication beyond what we know today. This will pose ethical challenges as well as technical ones”* (p. 33) + a dedicated outer space theme (p. 62).
 - [2024 Trends Report](#), which for the first time included trends on outer space (p. 26).
- [UN Summit of the Future in Sept 2024](#): world leaders adopted [Pact for the Future](#), with:
 - New, explicit mention of “exploration” in Action #56 on  *space*: *“We will strengthen international cooperation for the exploration and use of outer space for peaceful purposes to the benefit of all humanity”* (p. 39). #56 also “encourages the Committee on the Peaceful Uses of Outer Space to further consult on the proposal to hold a fourth [United Nations Conference on the Peaceful Exploration of Outer Space](#) (UNISPACE IV) in 2027”.
 - a [Declaration on Future Generations](#) with a commitment to act on *“Fostering an organizational culture that is future-oriented and mainstreamed across the United Nations system in order to facilitate science-and evidence-based decision-making by developing diverse capabilities, including anticipatory planning, foresight and futures literacy”* (p. 60). See [UN Futures Lab Network](#).
- UN Secretary-General recently formed a [Scientific Advisory Board](#):
 - 2024 statement on [Trust in Science](#), underscoring strategic foresight, need for public repositories of accessible/transparent/plain language summaries of scientific research; international panels on AI & disruptive tech to inform policies/governance

The SETI Committee should watch these developments, and engage, in the years ahead.

Protocols: a Draft Revised *Declaration* to discuss

- The Protocols task group (C. Oliver, K. Denning, L. Tennen, M. Garrett) has produced a Draft Revised “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)”, which Committee members have received. L. Tennen will review the changes in the next presentation, and then M. Garrett will discuss the process for feedback from the Committee and larger SETI research community. Your input is essential. We need your collegiality, patience, and expertise as we proceed.
- Two aspects of the Draft that we might discuss today? after L. Tennen presents:
 - The Preamble includes the idea of ‘**Best Practices Guidelines**’ to accompany the Declaration of Principles
 - Idea initially floated in [2019 paper](#): these could be less formal, produced by a small working group for each topic, and more frequently updated. E.g. Best Practices for: projects with students; for projects with citizen scientists; for data handling; for communicating about technosignatures research to broad audiences, etc.
 - **Post-Detection Sub-Committee**
 - Existed from ~1996-2014, but not for the past 10 years. Varying activities, remits, and ToR in the past – worth reviewing before we set new ToR and expectations
 - We should reflect on current overall Committee composition, who we need to add, and what a PD SubComm could really achieve. Some potential activities require expertise that we may not have at sufficient levels, and/or would realistically require funding. Much needs to be done, but the SETI Committee can’t realistically do it all.

Protocols: a note about history

- This history is generally under-appreciated. The initial formulation of the Protocols (1989) took the better part of a decade, and subsequent accompanying documents and revisions also took much time and effort.
- An update is not a trivial process, given the necessary review of previous work, plus the increasing complexity of technosignatures research, the expansion in the number of people doing technosignatures work, and the speed and volatility of today's communication landscape. (Recall that the first Protocols came before the WWW.)
- We need to already be looking ahead to the future  committee and the *next* update of the Declaration.
- We can't simplify the future for them. But we *can* make it easier for them to access and review the 60+ years of work already done on post-detection, Protocols etc. Some of it is at risk of being lost.
- Current Protocols revision work, and upcoming work by the IISL Working Group on SETI & Law and other projects, would be **greatly** facilitated by more easily accessible Committee-related PD history resources. These need to be collected, indexed, archived, and made available. L. Walton has been working hard on the history and deserves our thanks. But it's a big job, needs additional labour, and needs funding.

If you'd like to learn more about the history ... contact me at kdenning@yorku.ca for resources.